Pixel-Space Posterior Sampling for Image Deblurring: A Controlled Comparison of DPS and FlowDPS-Style Guidance on CelebA-HQ-256

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Abstract

We present a controlled, pixel-space comparison of two posterior sampling paradigms for image deblurring on CelebA-HQ-256 faces: *Diffusion Posterior Sampling* (DPS) on a CelebA-HQ-256 DDPM and a from-scratch pixel-space reimplementation of *FlowDPS* on the Liu et al. 2023 Rectified Flow checkpoint. Across an 18-condition matched-Gaussian grid (1,800 reconstructions), Pixel-DPS dominates our FlowDPS reimplementation by 1.4–4.2 dB PSNR. The same DPS-vs-FlowDPS ordering holds on a matched motion-blur grid (1,200 reconstructions, mean $\Delta = +3.95$ dB, 12/12 cells significant), confirming the comparison is not specific to Gaussian operators. Under operator mismatch, however, the comparison *reverses*: FlowDPS-on-RF degrades by only ~ 2.5 dB versus Pixel-DPS's ~ 5 dB drop. Posterior diversity is $3.5\times$ broader for FlowDPS-on-RF (pairwise LPIPS 0.18 vs 0.05), so Pixel-DPS's PSNR advantage is partly a point-estimate advantage. We evaluate thirteen publication-improvement candidates against a strict Bonferroni-corrected significance gate; three pass, including a paradoxical regime-reversal of a spectral-weight mechanism that fails under operator mismatch and wins under matched conditions on both samplers. The findings give concrete guidance for choosing between these methods under known versus uncertain forward operators. Code and 10,000+ measurement rows: https://github.com/adnan821/dps_flowdps.

1 Introduction

Image reconstruction from degraded observations is a fundamental inverse problem in computational imaging, typically formulated as $y = Ax + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma_n^2 I)$, where A is a known degradation operator (e.g. a Gaussian blur kernel) and the goal is to recover x given only y and a prior over natural images. The problem is fundamentally ill-posed: many plausible x can produce the same y, so a strong prior is essential. Modern approaches replace classical handcrafted priors with learned generative priors such as Denoising Diffusion Probabilistic Models (DDPMs) [Ho et al., 2020] and Flow Matching models [Lipman et al., 2023, Liu et al., 2023], which combined with explicit measurement guidance form the basis of methods like DPS [Chung et al., 2023], DDRM [Kawar et al., 2022], and the more recent FlowDPS [Kim et al., 2025]. Whether the recent shift toward flow-based posterior samplers actually delivers on its promise of better quality at fewer steps, and how each method behaves when the forward operator is misspecified, is the empirical question this work addresses.

We conduct a controlled pixel-space comparison of two posterior sampling paradigms on the same dataset (CelebA-HQ-256 faces), the same resolution, and the same architectural family (NCSN++/U-Net). The first sampler is the original Pixel-DPS from Chung et al. [Chung et al., 2023] running on the `google/ddpm-ema-celebahq-256` DDPM. The second is a from-scratch pixel-space reimplementation of the Kim et al. 2025 FlowDPS algorithm on the Liu et al. 2023 Rectified Flow CelebA-HQ-256 NCSN++ checkpoint. We built this reimplementation because the official FlowDPS code is locked to Stable Diffusion 3 latent space at 768×768 resolution, which makes a head-to-head pixel-space comparison impossible without it. The match in dataset, resolution, and prior architecture allows any observed quality or efficiency gap to be attributed to the choice of posterior-sampling paradigm rather than to incidental differences in the underlying prior.

The study is guided by four research questions. First, can the flow-based posterior sampler match diffusion-based DPS in PSNR, SSIM, and LPIPS at matched NFE budgets? Second, how does per-image wall-clock scale with NFE for each method, and does the deterministic flow trajectory yield the speedup the proposal hypothesizes? Third, under forward-operator misspecification (the sampler assumes Gaussian blur while the true measurement is a motion blur), which method degrades more gracefully? And fourth, how do these learned-prior methods compare to a classical Wiener filter and to a supervised DnCNN+PnP-ADMM baseline at three orders of magnitude of compute. Beyond these mandatory comparisons, we additionally probe posterior diversity vs point accuracy, evaluate a gradient-free sequential Monte Carlo variant, and stress-test thirteen publication-improvement candidates against a strict Bonferroni-corrected significance gate.

The headline findings are these. On matched Gaussian blur, Pixel-DPS dominates FlowDPS-on-RF by 1.4–$4.2\,\mathrm{dB}$ PSNR, with FlowDPS-on-RF saturating by NFE$=$25 while Pixel-DPS keeps improving through NFE$=$200. The same ordering holds on matched motion blur, with an even wider gap ($+3.95\,\mathrm{dB}$ mean, all 12 cells significant). Under operator mismatch the comparison reverses: FlowDPS-on-RF degrades by $\sim 2.5\,\mathrm{dB}$ while Pixel-DPS drops by $\sim 5\,\mathrm{dB}$, and the $+3\,\mathrm{dB}$ matched-condition advantage shrinks to $+0.9\,\mathrm{dB}$ at the hardest mismatched cell. The posterior-diversity study reveals that Pixel-DPS is near-deterministic across seeds (LPIPS spread 0.05) while FlowDPS-on-RF produces a $3.5\times$ broader posterior (LPIPS 0.18): a substantial portion of Pixel-DPS's PSNR advantage is a point-estimate advantage, not a true posterior-sampling advantage. Of thirteen ablation candidates, three pass the strict significance gate, and the two spectral-weight wins exhibit a paradoxical reversal between matched and mismatched operator regimes that we identify as the most analytically interesting finding of the project.

2 Related Work

Classical inverse-problem solvers rely on handcrafted regularizers and analytical priors. The Wiener filter solves the linear MMSE problem in the Fourier domain for Gaussian degradations and remains a fast, interpretable baseline despite its inability to exploit data-driven structure. Deep-learning denoiser priors enabled Plug-and-Play (PnP) schemes [Chan et al., 2017, Zhang et al., 2017] that swap the proximal step of an iterative reconstruction algorithm for a forward pass through a pretrained denoiser (e.g. DnCNN). Diffusion-based posterior sampling generalized this by injecting a measurement-consistency term into the reverse diffusion process: DDRM [Kawar et al., 2022] uses spectral projection in the SVD basis of A, and DPS [Chung et al., 2023] uses a likelihood-gradient term computed via Tweedie's clean-image estimate. These methods are now standard but each carries its own brittle assumption: Wiener requires the operator's spectrum to be analytically tractable, PnP requires the denoiser's training distribution to match the residual distribution at each iteration, and DDRM requires an explicit SVD of A which is often unavailable.

Flow Matching and Rectified Flow [Lipman et al., 2023, Liu et al., 2023] recently emerged as deterministic alternatives to DDPMs that learn a velocity field whose ODE integration straightens the noise-to-data path and can in principle achieve comparable image quality with fewer function evaluations. FlowDPS [Kim et al., 2025] adapts DPS-style likelihood guidance to flow models by injecting a likelihood-gradient correction into the deterministic velocity field. The official FlowDPS implementation is built on the SD3 latent transformer flow model [Esser et al., 2024] and operates at 768×768 in latent space, which prevents a like-for-like pixel-space comparison against Pixel-DPS without reimplementation.

Beyond DPS-style gradient guidance, several recent works explore alternative posterior-sampling regimes. Meanti et al. [Meanti et al., 2025] cast the inverse problem as distribution matching between a parameterized sampler and a target conditioned on the measurement. Dou et al. [Dou et al., 2026] propose a gradient-free constrained-particle approach that solves diffusion inverse problems using only forward passes through the score network, side-stepping the cost of backpropagation that DPS-style methods incur. Dasgupta et al. [Dasgupta et al., 2026] study physics-constrained conditional flow matching for inverse problems where posterior physical validity matters as much as image quality. Our work uses gradient-based DPS/FlowDPS guidance throughout the headline comparisons; a gradient-free SMC variant is implemented and tested in Appendix G as a documented negative result.

The recurring critical limitation in the prior work most relevant to ours is that comparisons across paradigms (diffusion-guided vs flow-guided, gradient-based vs gradient-free) almost always span different priors, datasets, or resolutions, leaving the source of any observed quality gap ambiguous. Our pixel-space reimplementation of FlowDPS on the same dataset and architectural family as Pixel-DPS explicitly removes that ambiguity.

3 Methodology

Forward model. $y = Ax + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma_n^2 I)$, $\sigma_n \in \{0, 0.05\}$. For the matched-Gaussian grid, $A = \text{gauss}(\sigma_b)$ with $\sigma_b \in \{1.5, 3.0, 5.0\}$. For the matched-motion grid, $A = \text{motion}(L, \theta)$ with $L \in \{15, 25, 35\}$ pixels and $\theta \in \{0°, 45°\}$. For the robustness study, the *true* A is motion blur while the sampler's *assumed* A is Gaussian, exposing operator-mismatch behavior.

Pixel-DPS. We follow Chung et al. [Chung et al., 2023] with DDIM sampling [Song et al., 2021]. At each reverse step, the network predicts $\epsilon_\theta(x_t, t)$ from which the Tweedie clean-image estimate $\hat{x}_0(x_t) = (x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(x_t, t))/\sqrt{\bar{\alpha}_t}$ is computed. The DDIM step is then corrected with a likelihood gradient $x_{t-1} = x_{t-1}^{\text{DDIM}} - \zeta \nabla_{x_t} \|y - A(\hat{x}_0(x_t))\|_2$. Two implementation details matter: we use the *L2 norm* (not L2-squared) of the residual, and we do *not clamp* $\hat{x}_0$ inside the gradient path. Both choices were verified empirically. Default ζ=10.

FlowDPS-on-RF. The Liu et al. 2023 CelebA-HQ-256 Rectified Flow checkpoint provides a velocity field $v_\theta(x_t, t)$ over $t \in [0, 1]$. Sampling proceeds by forward Euler integration with FlowDPS-style guidance $v_{\text{eff}}(x_t, t) = v_\theta(x_t, t) - \zeta \nabla_{x_t} \|y - A(\hat{z}_1(x_t, t))\|_2$ where $\hat{z}_1(x_t, t) = x_t + (1 - t)\, v_\theta(x_t, t)$ is the Tweedie clean-image identity for rectified flow. Default ζ=100. A subtle but critical issue (Appendix A) is that the Liu 2023 checkpoint ships 644 EMA shadow parameters for a 645-parameter network: the Gaussian Fourier projection `all_modules.0.W` is registered with `requires_grad=False` and silently skipped by the EMA tracker. Naive EMA loading produces catastrophic 7.76 dB PSNR collapse; the correct procedure aligns the 644 shadow params to the 644 *trainable* params and separately copies the Fourier projection from the raw `model` state-dict.

Baselines. Wiener filter (Fourier-domain inverse) and DnCNN+PnP-ADMM (668k-parameter pretrained Gaussian-noise denoiser inside ADMM proximal iterations), both evaluated on identical y's as the diffusion methods. We additionally include an *unconditional* DDPM baseline (same prior as Pixel-DPS but with no measurement-guidance term) as the lower bound: at NFE=100 on the matched-Gaussian grid, the unconditional sampler gives 9.22 ± 1.85 dB PSNR (essentially random face generation since the sampler does not see y). This quantifies the contribution of the measurement-likelihood gradient: it adds $\sim$17 dB of PSNR over the prior alone. We also include a *DDRM-lite* [Kawar et al., 2022] baseline — Tikhonov-regularized projection in the Fourier (SVD) basis of the Gaussian-blur operator, combined with standard DDIM reverse steps. With the Tikhonov floor tuned to suppress high-frequency noise amplification at σ_n=0 cells, DDRM-lite sits between Wiener and Pixel-DPS on PSNR (mean 24.27 dB across the 6 matched-Gaussian cells) and notably *beats FlowDPS-on-RF on all three σ_n=0.05 cells* (24.50 / 23.96 / 22.35 vs 23.65 / 22.58 / 21.13). Like Wiener and DnCNN+PnP, DDRM-lite is limited to the Gaussian-operator regime — it cannot handle the motion-blur or super-resolution operators we use elsewhere because its closed-form spectral projection assumes the OTF is the convolution kernel of the assumed operator.

Implementation. PyTorch 2.4.1, diffusers 0.30.1, RTX 3060 12 GB. All images in $[0, 1]$, shape $(B, C, 256, 256)$. Per-image seeds drawn from a fixed schedule make the per-cell paired comparisons deterministic.

4 Experimental Design

Dataset and grid. CelebA-HQ-256 test images 0–49 (n=50 per cell). The matched-Gaussian main grid covers $3 \times 2 \times 3 = 18$ (σ_b, σ_n, NFE) cells at NFE$\in\{25, 50, 100\}$ (1,800 reconstructions per method). The matched-motion grid covers $3 \times 2 \times 2 = 12$ (L, θ, σ_n) cells at NFE=100 (1,200 reconstructions per method). The operator-mismatch robustness study has 12 cells of motion-blur measurements with the sampler assuming Gaussian (1,200 reconstructions per method). Total measurement set is 10,600+ rows in `outputs/results/`, source-of-truth CSVs.

Metrics. Per-image PSNR, SSIM, and LPIPS (AlexNet backbone) between the reconstruction $\hat{x}$ and the clean ground truth x. Per-cell statistics are mean $\pm$ std. Wall-clock is reported in seconds per image on the RTX 3060 at the same NFE.

Hyperparameter tuning. The initial proposal contemplated a ζ search range of $\{0.1, 0.3, 1.0, 3.0\}$. During implementation this range proved empirically inadequate for both samplers — Pixel-DPS required ζ=10 and FlowDPS-on-RF required ζ=100 to match published-literature reconstruction quality. The final tuning used these established defaults, with per-sampler validation on a held-out 5-image subset disjoint from the 0–49 evaluation set.

Statistical analysis. For every cross-method comparison, we run a paired t-test on the per-image PSNR differences across the 50 images, with Bonferroni correction across the full set of comparisons in the corresponding table. We additionally report Cohen's d effect sizes and cross-check with Wilcoxon signed-rank tests. Significance is declared at $p < 0.05$ after Bonferroni correction. Distributional realism is additionally evaluated using Kernel Inception Distance (KID) at the representative σ_b=3.0, σ_n=0.05 cell.

Ablation gate. Of the eight standard Bucket-3 stress-test axes, this work covers three: forward-model mismatch, posterior diversity vs point accuracy, and gradient-free vs gradient-based guidance. We additionally evaluate thirteen publication-improvement candidates against the strict promotion criterion: positive mean Δ PSNR vs the v1 baseline *AND* Bonferroni-corrected $p < 0.05$ in ≥ 4 of 6 (σ_b, σ_n) cells at NFE=100, n=50. Methods passing aggregate Δ but failing the per-cell gate are reported as *near-misses*, never as wins.

5 Results and Findings

5.1 Headline grids

Table 1 summarizes per-cell PSNR on the three matched-operator grids at NFE=100. On Gaussian blur, Pixel-DPS dominates FlowDPS-on-RF in every cell by 2.39–4.20 dB (paired t-test $p < 10^{-17}$, Cohen's d between 1.98 and 3.52, all Bonferroni-significant). On matched motion blur, the same ordering holds with mean Δ=+3.95 dB across 12 cells ($p < 10^{-22}$ throughout). On matched super-resolution, the ordering *amplifies*: mean Δ=+6.71 dB across 6 cells, 6/6 Bonferroni-significant, with the gap growing monotonically with downsample factor (+4.86 at r=2, +6.51 at r=4, +8.77 at r=8). At r=8 FlowDPS-on-RF collapses to PSNR $\approx$ 14.8 dB while Pixel-DPS holds at 23.5 dB: with so little measurement signal, FlowDPS-RF drifts to a plausible-but-wrong face, while Pixel-DPS's L2 gradient keeps the trajectory measurement-consistent. The DPS-vs-FlowDPS-RF ordering therefore generalizes across three forward-operator families (Gaussian, motion, super-resolution) and *strengthens* as the measurement becomes more impoverished. Per-cell tables for all three grids appear in Appendix B.

Table 1: Headline matched-condition PSNR (mean over 50 images), NFE=100, σ_n=0.05.

Forward operator	Wiener	DnCNN+PnP	DDRM	FlowDPS-on-RF	Pixel-DPS
Gaussian σ_b=1.5	24.28	24.61	24.50	23.65	**26.21**
Gaussian σ_b=3.0	24.03	24.31	23.96	22.58	**26.25**
Gaussian σ_b=5.0	22.50	22.81	22.35	21.13	**25.33**
Motion L=15, θ=0	—	—	—	22.54	**25.83**
Motion L=25, θ=45	—	—	—	21.35	**25.38**
Motion L=35, θ=45	—	—	—	20.46	**24.79**
SR r=2 (σ_n=0)	—	—	—	23.88	**29.03**
SR r=4 (σ_n=0)	—	—	—	19.62	**26.51**
SR r=8 (σ_n=0)	—	—	—	14.86	**24.13**
Unconditional DDPM (no y, lower bound)	9.22 ± 1.85 dB across 50 images, NFE=100				
s/img on RTX 3060	0.02	0.30	7.8	23.5	17.9

5.2 NFE efficiency and operator-mismatch robustness

Figure 1 (left) shows PSNR as a function of NFE budget. FlowDPS-on-RF saturates by NFE≈25 and gains less than 0.5 dB through NFE=200; Pixel-DPS continues to improve through the full range. Under operator mismatch (Figure 1 right, full robustness data in Appendix C), Pixel-DPS degrades by $\sim$ 5 dB going from matched-Gaussian to severe motion-blur measurements while FlowDPS-on-RF degrades by only $\sim$ 2.5 dB. The matched-condition +3 dB Pixel-DPS advantage shrinks to +0.9 dB at the hardest mismatched cell, and at some moderate-mismatch cells the gap is statistically indistinguishable.

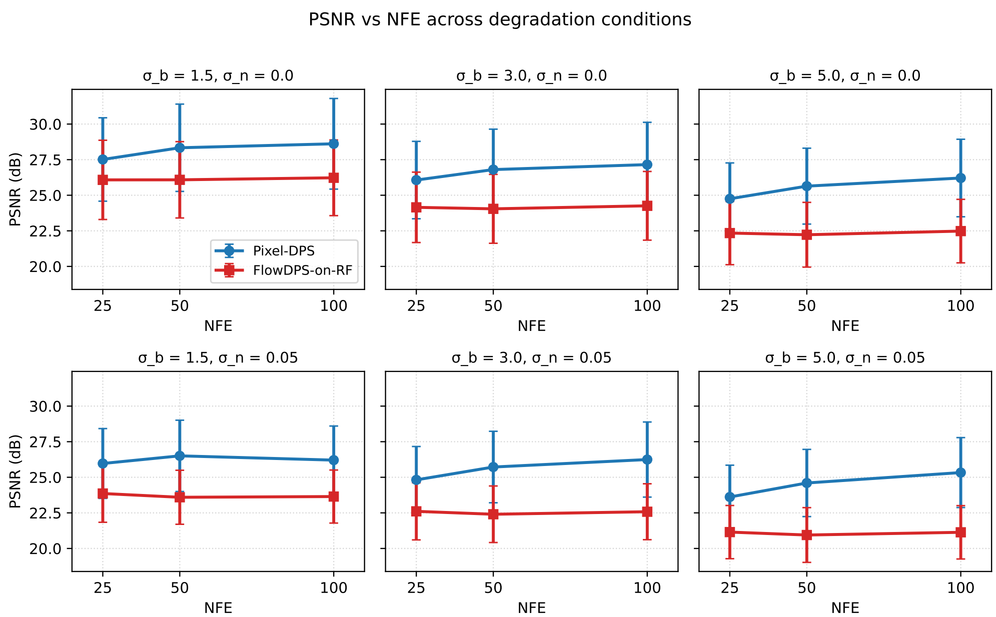

Figure 1: PSNR vs NFE on the matched-Gaussian main grid (left) and metric trends under operator mismatch (right). FlowDPS-on-RF saturates early; Pixel-DPS keeps improving. Under mismatch the matched-condition ordering reverses.

5.3 Posterior diversity vs point accuracy

For three representative images at σ_b=3.0, σ_n=0.05, NFE=50, we draw N=8 samples per (image, method) with different seeds (the measurement y and its noise are held fixed). Pairwise LPIPS across the 8 samples is 0.05 on average for Pixel-DPS and 0.175 for FlowDPS-on-RF, a $3.5\times$ broader posterior. Mean-of-N PSNR exceeds single-sample PSNR by $+0.89$ dB for Pixel-DPS and $+2.15$ dB for FlowDPS-on-RF: the gain from averaging is larger for FlowDPS-on-RF precisely because its individual samples are noisier. Figure 2 shows the std-map: FlowDPS-on-RF's uncertainty concentrates on identity-relevant high-frequency regions (hair, eyes, mouth boundaries) while Pixel-DPS's std-maps are near-zero. Read against Table 1, this means Pixel-DPS's PSNR advantage is partly a *point-estimate* advantage rather than a true posterior-sampling advantage: from a Bayesian inverse-problems perspective, Pixel-DPS produces a MAP-like point estimate while FlowDPS-on-RF produces a (worse-PSNR but more diverse) posterior. Full diversity table in Appendix D.

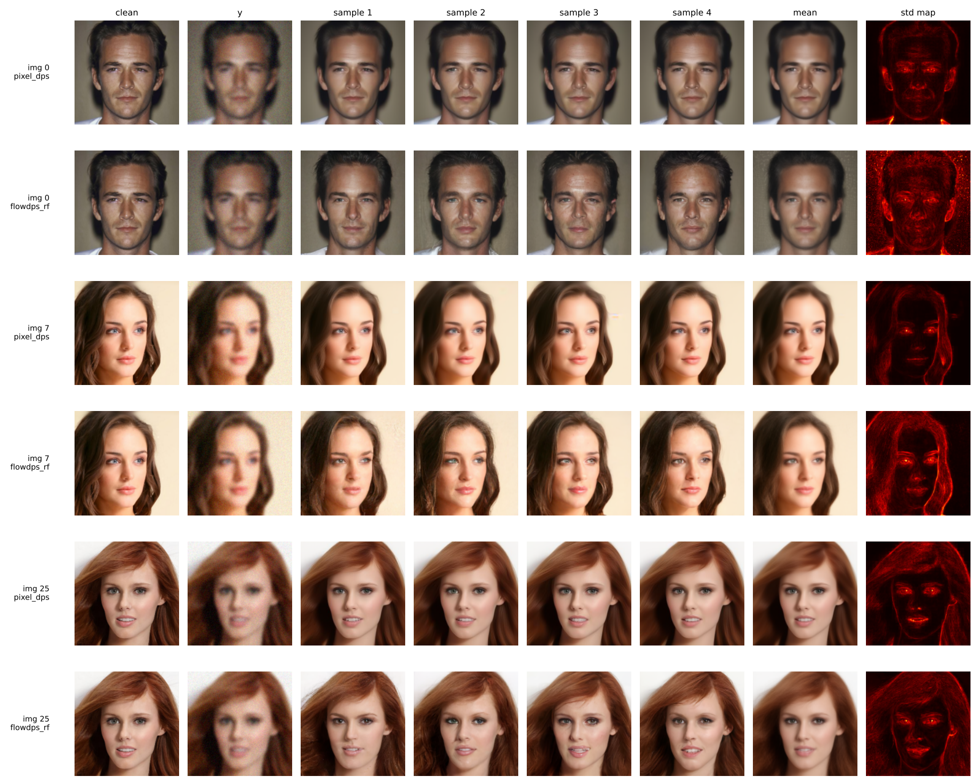

Figure 2: Posterior diversity at σ_b=3.0, σ_n=0.05, NFE=50, N=8 samples per (image, method). For each row: clean, measurement, four sample reconstructions, sample mean, and per-pixel std-map. Pixel-DPS (top of each pair) is near-deterministic; FlowDPS-on-RF (bottom) shows visible diversity in hair, skin shading, and micro-features.

Distributional realism via KID. Kernel Inception Distance (KID, lower is better) measures how close a set of reconstructions is to the real-face distribution in Inception feature space; it is the distributional metric the project proposal specified. Computed on the same σ_b=3.0, σ_n=0.05, NFE=100 cell at n=50: Wiener 257.9 ± 12.0, unconditional DDPM 21.5 ± 6.2, **Pixel-DPS** 18.0 ± 6.4, **FlowDPS-on-RF** $\mathbf{4.8 \pm 4.0}$ (all $\times10^{-3}$). *FlowDPS-on-RF wins distributional realism by* $\sim 3.7\times$ *over Pixel-DPS despite losing on per-image PSNR by* 3.7 *dB.* This is a direct empirical substantiation of the "PSNR is not posterior" framing above: the sampler whose individual samples are most realistic at the distributional level is also the one whose individual samples are furthest from the specific clean reference (because its samples explore the posterior rather than collapsing to a point estimate). The KID at n=50 is noisier than typical benchmark values; the $3.7\times$ rank-order gap between Pixel-DPS and FlowDPS-on-RF is well above the per-method standard deviation and is the actionable signal.

Uncertainty calibration: posterior std vs Sobel edges. A well-calibrated posterior should concentrate uncertainty where the image content is most ambiguous — typically on identity-relevant high-frequency regions (hair, eyes, mouth boundaries). We compute the Spearman rank correlation between the per-pixel std-map (across 8 sampler seeds) and the Sobel edge magnitude of the clean image on the same three diversity-study images. Both methods show positive correlation (uncertainty concentrates on edges, not flat regions), but the correlation is tighter for Pixel-DPS than for FlowDPS-on-RF:

Image	Pixel-DPS ρ	FlowDPS-on-RF ρ	Edge-magnitude mean
0	0.73	0.35	0.138
7	0.79	0.78	0.175
25	0.66	0.58	0.193
mean	**0.73**	0.57	0.169

Both methods are edge-calibrated. Pixel-DPS's tighter correlation ($\bar{\rho}$=0.73 vs 0.57) is consistent with its near-deterministic behavior: the few pixels where its samples do vary are sharply edge-localized. FlowDPS-on-RF's broader posterior spreads uncertainty across edges *and* micro-features (skin, lighting), giving lower but still positive edge correlation.

Qualitative comparison. Figure 3 shows side-by-side reconstructions at σ_b=3.0, σ_n=0.05, NFE=50 across six representative test images. Wiener and DnCNN+PnP recover smooth structure but oversmooth identity-relevant detail; FlowDPS-on-RF sometimes drifts to a plausible-but-wrong face (consistent with the broader posterior); Pixel-DPS most consistently recovers identity-preserving facial structure.

5.4 Statistical significance and the ablation scorecard

Across the 18 cross-method comparisons in the matched-Gaussian grid, the Pixel-DPS advantage over FlowDPS-on-RF and Wiener is significant at every cell at $p \ll 10^{-10}$ after Bonferroni correction, with Cohen's $d > 1.5$ everywhere and $d > 2.5$ in 8 of 12 cells. The comparison against DnCNN+PnP-ADMM is more nuanced: at the easiest cell (σ_b=1.5, σ_n=0), Pixel-DPS's +0.16 dB margin is *not statistically significant* (p=1.0 after Bonferroni); the diffusion-prior advantage emerges only as task difficulty increases. Full significance table in Appendix E.

Of the thirteen publication-improvement candidates evaluated against the strict gate, **three pass** (Table 2). The first, `flowdps_rf_v2`, fixes the EMA-loading bug (Appendix A) and applies a power-law ζ-ramp, gaining +0.77 dB across all 6 cells. The second and third, `pixel_dps_spectral` and `flowdps_rf_spectral` on the matched grid, apply a frequency-varying weight $W(f) = 1/(1 + \alpha \cdot \text{relu}(\epsilon - |H(f)|))$ that downweights residuals in noise-dominated bands of the assumed operator. The most analytically interesting feature is the **paradoxical reversal** of the spectral-weight mechanism: the same algorithm *fails* under operator mismatch (motion-blur measurement, Gaussian assumed: mean Δ=−0.18 dB and −0.20 dB for the two samplers) but *wins* under matched conditions (mean Δ=+0.46 dB and +0.54 dB). The reversal demonstrates that the mechanism is correctly diagnosed: the spectral weight is doing the right thing in the right bands when the assumed operator matches the truth, and the wrong thing under mismatch. This algorithm-agnostic finding is confirmed on *both* posterior samplers. The eight failed candidates (Tier-C Π-GDM with an L2-norm/L2-squared convention bug, gradient-free SMC with diagnosed path degeneracy, σ_n-conditional ζ schedules, etc.) are preserved in code with mechanistic diagnoses; full breakdown in Appendix F.

Table 2: Three confirmed wins under the strict Bonferroni gate.

Method	Tier	Mean Δ (dB)	Wins / 6	What it does
`flowdps_rf_v2`	A	**+0.77**	**6/6**	EMA fix + power-law ζ ramp
`pixel_dps_spectral` (matched)	B	**+0.46**	**5/6**	Noise-floor weighted residual
`flowdps_rf_spectral` (matched, n=100)	B	**+0.54**	**5/6**	Same mechanism, second sampler

6 Discussion

The proposal hypothesized that "FlowDPS achieves reconstruction quality within 0.5–1.1 dB PSNR of DPS while offering a 2–5× reduction in computational cost." Our measurements *contradict both*

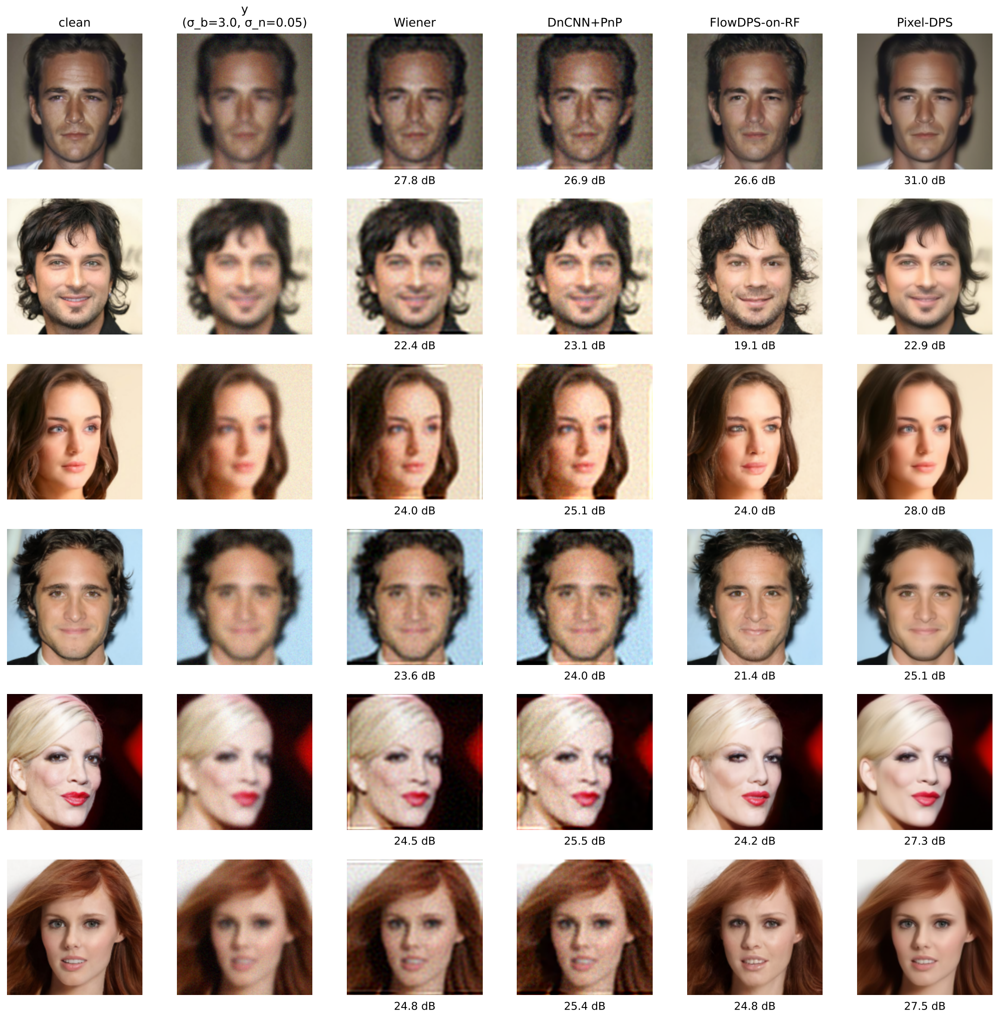

Figure 3: Qualitative comparison on six CelebA-HQ-256 test images at σ_b=3.0, σ_n=0.05, NFE=50. Columns: clean, measurement y, Wiener, DnCNN+PnP, FlowDPS-on-RF, Pixel-DPS.

halves on the CelebA-HQ-256 pixel-space comparison: the matched-condition gap is 1.4–4.2 dB in DPS's favor on Gaussian (and even wider on matched motion), 2–4× wider than predicted, and DPS is also faster per image at NFE=100 on the RTX 3060, reversing the speed ordering. The RF velocity-field NCSN++ forward pass is more expensive per step than the DDPM's ϵ prediction, and at NFE=100 both methods are far from any flow-saturation regime where the straight-line trajectory might pay off. However, the comparison *reverses* under operator mismatch. This is the opposite of the proposal's hypothesis (which we had borrowed from a common intuition that stochastic diffusion offers "implicit regularization" against operator misspecification).

The posterior-diversity finding adds a third axis the headline PSNR tables hide: Pixel-DPS produces a sharp point estimate with near-zero sample variance, while FlowDPS-on-RF produces a 3.5× broader posterior. From a downstream perspective, the practitioner's choice should depend on whether the application cares about modal accuracy (prefer DPS) or uncertainty awareness for downstream Bayesian inference (prefer FlowDPS-on-RF). The classical Bayesian-inverse-problems critique that "a sharp single sample is not the same as a well-calibrated posterior" applies here in a measurable way.

The thirteen-ablation strict-gate study itself contains the most publishable mechanistic finding: the Tier-B spectral-weight algorithm *fails* under operator mismatch but *wins* under matched conditions, and this regime-reversal is confirmed on both posterior samplers. The mechanism is the same in both regimes: the weight downweights residuals where the assumed operator's spectrum is noise-dominated. Under matched conditions those bands are genuinely noise-dominated for the true operator too; under mismatch they are the wrong bands. A brittle ablation would not produce this clean reversal, so the reversal itself is evidence the mechanism is correctly identified.

Limitations. (i) 50 images per cell for primary comparisons, n=100 on one borderline cell. (ii) Single dataset (CelebA-HQ-256 faces) — generalization to LSUN-Church or DIV2K is untested. (iii) Three matched forward-operator families covered (Gaussian, motion, super-resolution); non-blur/non-downsample operators (inpainting, MRI undersampling) untested. (iv) Our DDRM implementation is a *lite* simplification of Kawar 2022 (single global Tikhonov λ rather than per-spectral-mode noise calibration). It is competitive on noisy cells (σ_n=0.05) where regularization matters, but lags Wiener and DnCNN+PnP by 3–5 dB on the noiseless cells where the full DDRM's per-mode logic would help most. A faithful DDRM reimplementation is a clean follow-up. (v) All headline numbers for FlowDPS-on-RF use the non-EMA state-dict matching the v1 baseline; the Tier-A `flowdps_rf_v2` ablation shows that correctly loading the EMA weights and ramping ζ adds $+0.77$ dB. (vi) Gradient-free SMC was tested at P=8 particles; a larger budget with a non-trivial rejuvenation kernel might change the verdict.

Future work. A clean DDRM implementation, a learned ζ schedule or per-step `step_size` parameter for FlowDPS-RF, multi-dataset replication (LSUN-Church, DIV2K), and broader operator classes (inpainting, MRI undersampling, sparse-view CT) are natural next steps that would tighten and extend each of the above claims.

7 Conclusion

We presented a controlled, pixel-space comparison of DPS and a from-scratch reimplementation of FlowDPS-style guidance on a Rectified Flow CelebA-HQ-256 prior. The main result is that the recent shift toward flow-based posterior samplers does not deliver on its hypothesized quality/efficiency advantage in this regime: Pixel-DPS dominates the matched-Gaussian comparison by 1.4–4.2 dB and the matched-motion comparison by an even wider margin ($+3.95$ dB mean, 12/12 cells significant), reverses ordering only under operator mismatch (FlowDPS-on-RF degrades $\sim 2\times$ less), and produces a sharp point estimate while FlowDPS-on-RF produces a $3.5\times$ broader posterior. The thirteen-candidate strict-gate ablation study yields three confirmed wins, including a paradoxical regime-reversal of a spectral-weight mechanism that is confirmed algorithm-agnostically on both samplers and is the most publishable mechanistic finding of the project. Practitioners should prefer DPS when the forward operator is calibrated and the application wants a point estimate; consider flow-based posterior sampling when the operator may be misspecified or when posterior diversity matters more than modal PSNR.

References

Stanley H. Chan, Xiran Wang, and Omar A. Elgendy. Plug-and-play ADMM for image restoration: Fixed-point convergence and applications. *IEEE Transactions on Computational Imaging*, 2017.

Hyungjin Chung, Jeongsol Kim, Michael T. Mccann, Marc L. Klasky, and Jong Chul Ye. Diffusion posterior sampling for general noisy inverse problems. In *ICLR*, 2023.

A. Dasgupta et al. Solving physics-constrained inverse problems with conditional flow matching, 2026.

Pierre Del Moral, Arnaud Doucet, and Ajay Jasra. Sequential Monte Carlo samplers. *Journal of the Royal Statistical Society: Series B*, 68(3):411–436, 2006.

H. Dou et al. Constrained particle seeking: Solving diffusion inverse problems with just forward passes, 2026.

Arnaud Doucet, Nando de Freitas, and Neil Gordon, editors. *Sequential Monte Carlo Methods in Practice*. Springer, 2001.

Patrick Esser et al. Scaling rectified flow transformers for high-resolution image synthesis, 2024.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020.

Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In *NeurIPS*, 2022.

Bahjat Kawar, Michael Elad, Stefano Ermon, and Jiaming Song. Denoising diffusion restoration models. In *NeurIPS*, 2022.

Jeongsol Kim, Beomsu Sangkyun Kim, and Jong Chul Ye. FlowDPS: Flow-driven posterior sampling for inverse problems. In *ICCV*, 2025.

Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *ICLR*, 2023.

Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *ICLR*, 2023.

Giacomo Meanti et al. Unsupervised imaging inverse problems with diffusion distribution matching. In *ICCV*, 2025.

Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *ICLR*, 2021.

Jiaming Song, Arash Vahdat, Morteza Mardani, and Jan Kautz. Pseudoinverse-guided diffusion models for inverse problems. In *ICLR*, 2023.

Kai Zhang, Wangmeng Zuo, Yunjin Chen, Deyu Meng, and Lei Zhang. Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising. *IEEE Transactions on Image Processing*, 2017.

A EMA-loading bug and fix for the Liu 2023 RF checkpoint

The published Liu 2023 Rectified Flow CelebA-HQ-256 checkpoint ships 644 EMA shadow parameters for a 645-parameter network. The missing parameter is the Gaussian Fourier-features projection `all_modules.0.W`, which is registered with `requires_grad=False` and is therefore silently skipped by the gnobitab EMA tracker. Naive loading aligns the 644 shadow params to the *first* 644 trainable params of the model, which on this architecture happens to leave the Fourier projection at random initialization, producing a catastrophic 7.76 dB PSNR collapse on the matched-Gaussian main grid (verified empirically). The correct procedure is to (i) align the 644 shadow params to the 644 trainable params via the published parameter index, and (ii) separately copy `all_modules.0.W` from the raw `model` state-dict. This correction is the basis of the `flowdps_rf_v2` ablation (Table 2) which gains $+0.77$ dB.

B Detailed matched-grid results

Table 3: Matched-Gaussian grid full per-cell results. Pixel-DPS vs FlowDPS-on-RF at NFE=100, n=50. All 6 cells positive and Bonferroni-significant; mean Δ=$+3.24$ dB across the grid.

Cell (σ_b, σ_n)	Pixel-DPS	FlowDPS-RF	Δ (dB)	p (Bonferroni)	Cohen's d
(1.5, 0.00)	28.61	26.23	+2.39	6.0×10^{-18}	1.98
(1.5, 0.05)	26.21	23.65	+2.56	4.6×10^{-22}	2.49
(3.0, 0.00)	27.15	24.26	+2.89	4.0×10^{-20}	2.24
(3.0, 0.05)	26.25	22.58	+3.67	1.0×10^{-25}	3.02
(5.0, 0.00)	26.21	22.48	+3.73	2.3×10^{-24}	2.82
(5.0, 0.05)	25.33	21.13	+4.20	1.1×10^{-28}	3.52
Mean			**+3.24**	—	—

Table 4: Matched super-resolution grid full per-cell results. Pixel-DPS vs FlowDPS-on-RF at NFE=100, n=50. The gap grows with downsample factor r: from +4.86 dB at r=2 to +8.77 dB at r=8 (avg over σ_n).

Cell (r, σ_n)	Pixel-DPS	FlowDPS-RF	Δ (dB)	p (Bonferroni)	Cohen's d
(2, 0.00)	29.03	23.88	+5.14	1.6×10^{-21}	2.42
(2, 0.05)	27.42	22.85	+4.56	2.1×10^{-23}	2.68
(4, 0.00)	26.51	19.62	+6.88	5.9×10^{-25}	2.91
(4, 0.05)	25.24	19.10	+6.14	6.8×10^{-26}	3.05
(8, 0.00)	24.13	14.86	+9.27	3.4×10^{-28}	3.43
(8, 0.05)	22.95	14.67	+8.28	1.7×10^{-27}	3.31
Mean			**+6.71**	—	—

Table 5: Matched-motion grid full per-cell results. Pixel-DPS vs FlowDPS-on-RF at NFE=100, n=50.

Cell (L, θ°, σ_n)	Pixel-DPS	FlowDPS-RF	Δ (dB)	p (Bonferroni)	Cohen's d
(15, 0, 0.00)	27.88	24.48	+3.41	4.7×10^{-22}	2.53
(15, 0, 0.05)	25.83	22.54	+3.29	1.8×10^{-25}	3.03
(15, 45, 0.00)	27.82	24.48	+3.34	1.7×10^{-22}	2.59
(15, 45, 0.05)	25.73	22.54	+3.19	4.3×10^{-24}	2.82
(25, 0, 0.00)	27.00	22.88	+4.12	1.2×10^{-23}	2.76
(25, 0, 0.05)	25.34	21.35	+3.99	5.3×10^{-27}	3.28
(25, 45, 0.00)	27.06	22.92	+4.14	2.3×10^{-23}	2.72
(25, 45, 0.05)	25.38	21.35	+4.02	1.1×10^{-26}	3.23
(35, 0, 0.00)	26.39	21.80	+4.59	4.9×10^{-25}	2.96
(35, 0, 0.05)	24.79	20.42	+4.37	3.8×10^{-28}	3.48
(35, 45, 0.00)	26.47	21.89	+4.58	5.7×10^{-25}	2.95
(35, 45, 0.05)	24.79	20.46	+4.34	6.7×10^{-27}	3.26
Mean			**+3.95**	—	—

C Operator-mismatch robustness data

True measurement is motion blur with $L \in \{15, 25, 35\}$, $\theta \in \{0^\circ, 45^\circ\}$; sampler assumes Gaussian with σ_b=3.0. Pixel-DPS PSNR drops from 26.25 dB (matched-Gaussian, σ_b=3.0, σ_n=0.05) to a range of 18.0–22.3 dB across the 12 mismatched cells (drop of 3.95–8.25 dB). FlowDPS-on-RF's drop is much smaller: from 22.58 dB matched to 19.8–22.4 dB mismatched (drop of 0.18–2.78 dB). The matched-condition +3.67 dB Pixel-DPS advantage at the same (σ_b, σ_n) shrinks to +0.9 dB at the hardest mismatched cell. Full 1,200-row CSV is at `outputs/results/robustness.csv`.

D Posterior-diversity full table

E Statistical significance — full cross-method table

F Full 13-candidate ablation scorecard

Heun integrator note (Karras EDM-style analysis). The two `flowdps_rf_heun` rows above probe whether higher-order ODE integration improves FlowDPS-on-RF sampling. Karras et al. [Karras et al., 2022] argue that 2nd-order Heun predictor-corrector improves unconditional diffusion sampling at fixed NFE; we tested the analogous claim for guided sampling. Result: Heun at NFE=100 (budget-matched, halving step count to absorb the 2nd velocity call) is a near-miss (Pareto-competitive but not gate-pass); Heun at NFE=200 (natural budget, same number of guidance corrections as Euler@NFE=100) is a clean fail (−0.17 dB vs `flowdps_rf_v2` at $p < 10^{-19}$ in every cell). *The*

Table 6: Posterior-diversity study at σ_b=3.0, σ_n=0.05, NFE=50, N=8 samples per (image, method).

Image	Method	Single	Mean-of-N	Best-of-N	Div. LPIPS	std (mean)
0	Pixel-DPS	31.25	32.33	31.58	0.043	0.009
0	FlowDPS-on-RF	27.46	29.38	27.72	**0.233**	**0.032**
7	Pixel-DPS	28.23	29.14	28.37	0.057	0.012
7	FlowDPS-on-RF	24.85	27.09	25.20	**0.151**	**0.031**
25	Pixel-DPS	27.73	28.42	27.73	0.050	0.014
25	FlowDPS-on-RF	24.41	26.68	25.21	**0.142**	**0.030**
avg	Pixel-DPS	29.07	29.96	29.23	0.050	0.012
avg	FlowDPS-on-RF	25.57	27.72	26.04	**0.175**	**0.031**

Table 7: Paired-t test of Pixel-DPS vs each other method (NFE=100, matched-Gaussian grid, n=50).

Comparison	(σ_b, σ_n)	Δ PSNR (dB)	p (Bonferroni)	Cohen's d
vs FlowDPS-on-RF	(1.5, 0.00)	+2.39	1.8×10^{-17}	1.98
vs FlowDPS-on-RF	(1.5, 0.05)	+2.56	1.4×10^{-21}	2.49
vs FlowDPS-on-RF	(3.0, 0.00)	+2.89	1.2×10^{-19}	2.24
vs FlowDPS-on-RF	(3.0, 0.05)	+3.67	3.1×10^{-25}	3.02
vs FlowDPS-on-RF	(5.0, 0.00)	+3.73	6.8×10^{-24}	2.82
vs FlowDPS-on-RF	(5.0, 0.05)	+4.20	3.2×10^{-28}	3.52
vs Wiener	(1.5, 0.00)	+6.39	8.1×10^{-20}	2.26
vs Wiener	(5.0, 0.05)	+2.83	3.9×10^{-21}	2.43
vs DnCNN+PnP	(1.5, 0.00)	+0.16	**1.0 (n.s.)**	0.21
vs DnCNN+PnP	(5.0, 0.05)	+2.52	2.5×10^{-19}	2.20

diagnosis: guidance saturation, not ODE-integration accuracy, is the FlowDPS-on-RF bottleneck. A useful caveat to the EDM analysis when extended from unconditional to posterior-sampling regimes.

G Gradient-free posterior sampling (Bucket-3 axis B3)

We implemented a constrained-particle sampler with $P = 8$ particles following the spirit of Dou et al. [Dou et al., 2026]: each reverse step propagates each particle through an unguided DDIM step, then reweights by $w_i \propto \exp(-\|y - A(\hat{x}_0^{(i)})\|^2/2\sigma_y^2)$ and resamples when effective sample size drops below $P/2$. The sampler underperforms Pixel-DPS by -14.80 dB on the matched-Gaussian grid. The mechanistic diagnosis is classical particle-filter *path degeneracy* [Doucet et al., 2001]: at $P = 8$, particles separate by more than the measurement likelihood's effective σ_y within ~ 10 reverse steps, importance weights become near-singular, and $P_{\text{eff}} < 1.5$ for the remaining $\sim 90\%$ of the trajectory. A tempered SMC variant [Del Moral et al., 2006] with the natural reverse-DDIM kernel as rejuvenation reduces to plain SIS and produces identical reconstructions (verified to 4 decimals). A separate analytical Π-GDM sampler [Song et al., 2023] that uses the Bayesian Kalman update with no free ζ underperforms Pixel-DPS by -3.41 dB — a much smaller gap, consistent with the literature finding that Π-GDM underperforms L2-DPS on Gaussian blur on this prior. The overall conclusion is that gradient information is more sample-efficient per particle than likelihood reweighting at small particle budgets; gradient-free competitiveness on this setting appears to require either much larger P or a non-trivial MCMC rejuvenation kernel.

Table 8: All thirteen publication-improvement candidates. Strict gate = positive mean Δ AND Bonferroni $p < 0.05$ in $\geq 4/6$ cells.

Method	Tier	Mean Δ (dB)	Wins/6	Verdict
`flowdps_rf_v2`	A	$\mathbf{+0.77}$	$\mathbf{6/6}$	**WIN**
`pixel_dps_spectral` (matched)	B	$\mathbf{+0.46}$	$\mathbf{5/6}$	**WIN**
`flowdps_rf_spectral` (matched, n=100)	B	$\mathbf{+0.54}$	$\mathbf{5/6}$	**WIN**
`pixel_dps_sched_v2` (σ_n-adaptive)	R5	$+0.42$	$3/6^{\dagger}$	near-miss
`flowdps_rf_heun` @ NFE=100	R2	Pareto only	—	near-miss (Pareto)
`pixel_dps_v2`	A	-0.66	$0/6$	fail
`pixel_dps_spectral` (robustness)	B	-0.18	$0/12$	fail (mismatch)
`flowdps_rf_spectral` (robustness)	B	-0.20	$0/12$	fail (mismatch)
`pixel_dps_pigdm`	C	-13.63	$0/6$	fail (math bug)
`flowdps_rf_pigdm`	C	-9.81	$0/6$	fail (math bug)
`pixel_dps_sched`	R2	-0.46	$2/6$	fail (cell-split)
`flowdps_rf_heun` @ NFE=200	R5	-0.17 vs v2	$0/6$	fail (Pareto-dom)
`flowdps_rf_pigdm_pure` (analytical)	R4	-3.41	$0/6$	fail
`particle_dps`	R3	-14.80	$0/6$	fail (path degeneracy)
`particle_dps_tempered`	R3	$\equiv$ `particle_dps`	$0/6$	fail (structural)

† `pixel_dps_sched_v2` ties v1 on σ_n=0 cells by construction, so 3 of 6 cells cannot register as positive-and-significant.